Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

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EXPERIMENT 1

Analyze the trend of data science job postings over the last decade

Experiment 1.a: Analyze the trend of data science job postings over the last decade

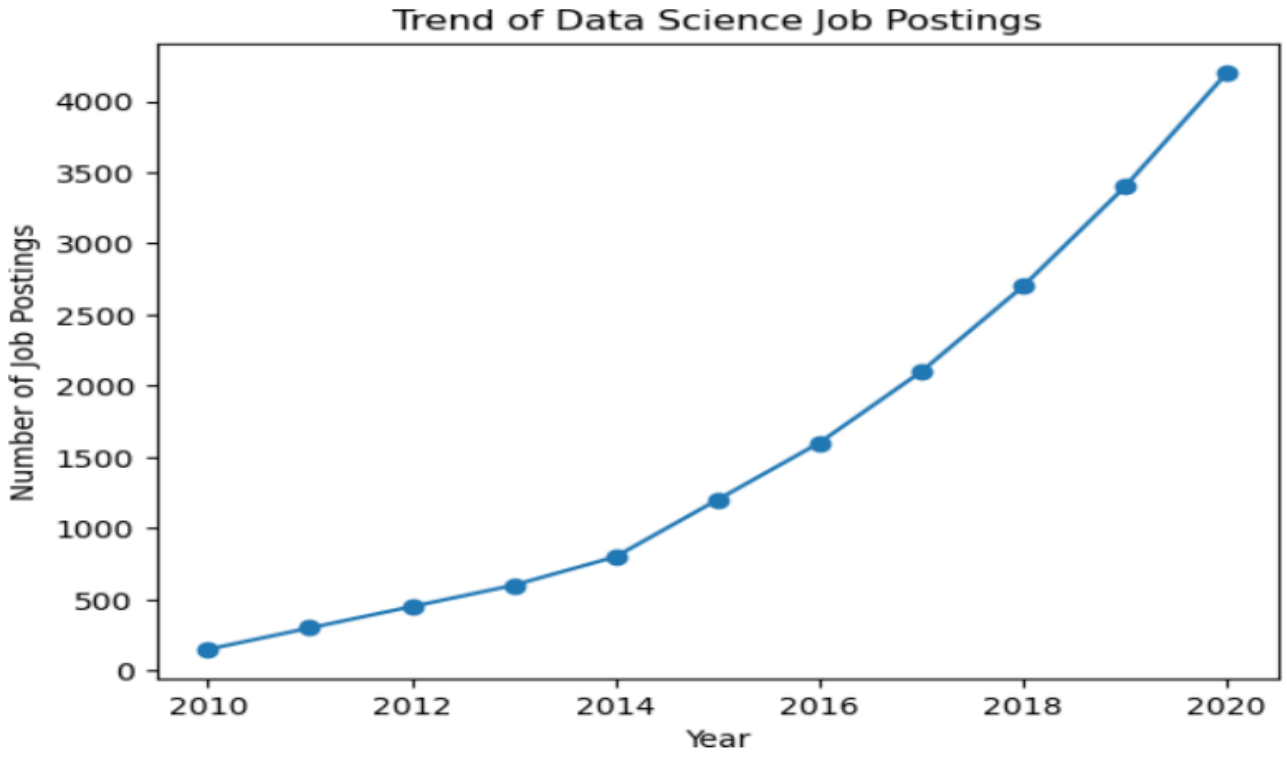
Aim

To analyze the trend in data science job postings over the last decade using data visualization.

Source Code

```
import pandas as pd
import matplotlib.pyplot as plt
data = {'Year': list(range(2010, 2021)),
'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}
df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job Postings'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
Year =2010, 2021
```

Output - Experiment 1.a



Result - Experiment 1.a

The trend of data science job postings was analyzed and visualized.

Experiment 1.b: Analyze and visualize the distribution of data science roles

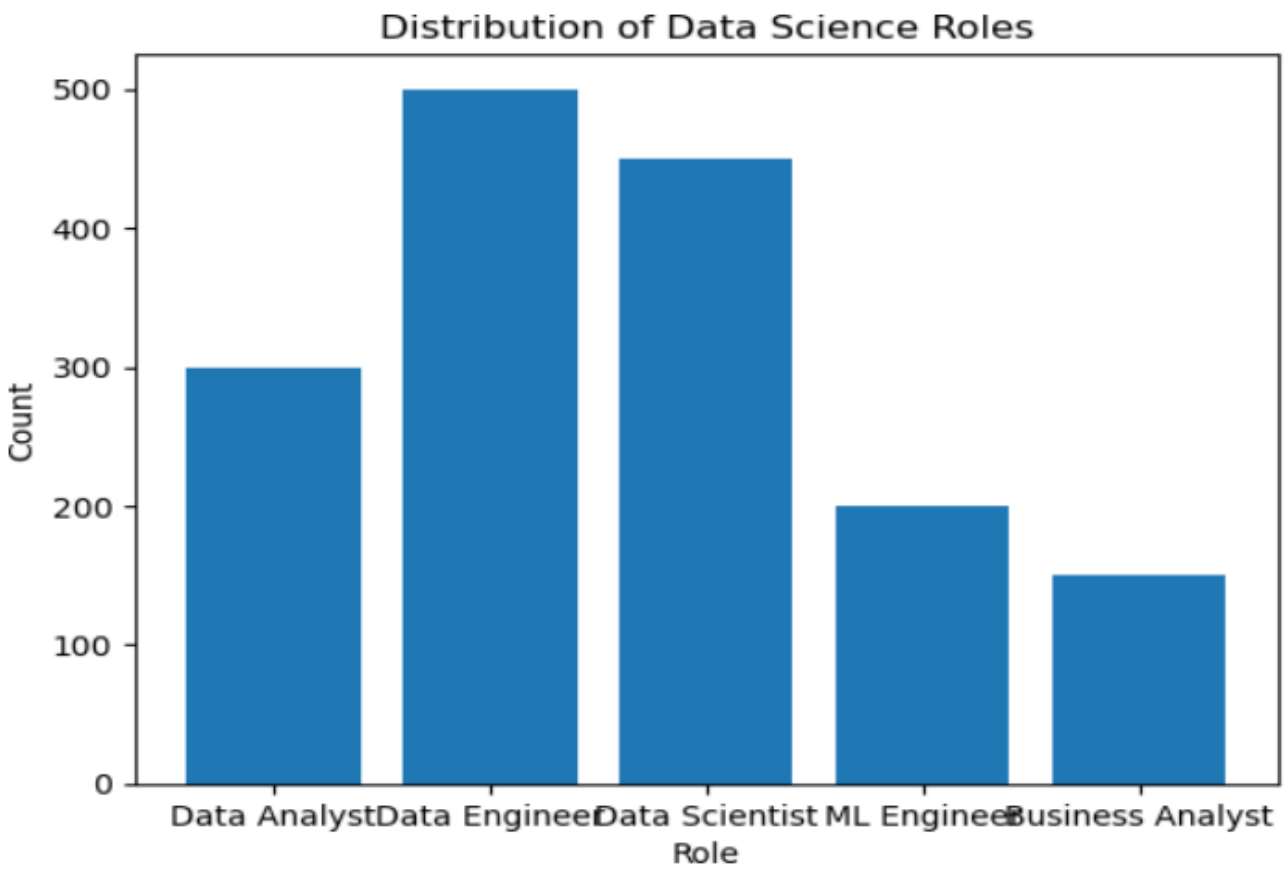
Aim

To analyze and visualize the distribution of data science roles using charts.

Source Code

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',
'Business Analyst']
counts = [300, 500, 450, 200, 150]
plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
roles = Data Analyst, Data Engineer, Data Scientist, ML Engineer, Business Analyst.
counts = 300, 500, 450, 200, 150.
```

Output - Experiment 1.b



Result - Experiment 1.b

The distribution of data science roles was categorized and visualized.

Experiment 1.c: Differentiate structured, unstructured and semi-structured data

Aim

To distinguish structured, unstructured, and semi-structured data using examples.

Source Code

```
structured_data = pd.DataFrame({
'ID': [1, 2, 3],
'Name': ['Alice', 'Bob', 'Charlie'],
'Age': [25, 30, 35]
})
print("Structured Data:\n", structured_data)
unstructured_data = "This is an example of unstructured data. It can be a piece of text, an image,
print("\nUnstructured Data:\n", unstructured_data)
semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
print("\nSemi-structured Data:\n", semi_structured_data)
```

Output - Experiment 1.c

Structured Data:
ID Name Age
0 1 Alice 25
1 2 Bob 30
2 3 Charlie 35
Unstructured Data:
This is an example of unstructured data. It can be a piece of text, an image, or a video file.
Semi-structured Data:
{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}

Result - Experiment 1.c

Structured, unstructured and semi-structured data were differentiated.

Experiment 1.d: Encrypt and decrypt given sensitive data

Aim

To encrypt and decrypt the given sensitive data.

Source Code

```
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
f.decrypt(token)
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

Output - Experiment 1.d

Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABmmnHEhy1ZC1-Dox-URNAvYqFFGfJemMb_fKmXqhATe3AAn
XEOnSzaNRINI4zAbNK8lgx2pprrNn-j6agbWDf30OwbrZsBuz8fnDdX7N35WwloDu3m6pVv
kwv8GYDSI8Fov1tV'
Decrypted Data: b'Rajalakshmi Engineering College.'
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Result - Experiment 1.d

The given sensitive data was encrypted and decrypted.